Project Overview

My dataset comes from a Kaggle Valorant Champions Tour dataset: https://www.kaggle.com/datasets/ryanluong1/valorant-champion-tour-2021-2023-data, and I am using only the 2026 VCT data for Kickoff and Masters Santiago. The dataset includes 21 CSV files with match results, player statistics, agent pick rates/bans, rounds, economy, and map information. There are roughly 50,000 to 100,000 total observations across all files, with about 20 to 30 predictors in the main tables. The variables include numeric variables such as ACS, K/D, and pick rate, as well as categorical variables such as map, agent, and tournament stage. Missing data is mainly common in areas such as lack of agent use. For example, reyna is a very unpopular agent, so I plan to null this value in the pick-rate column as it is so insignificant.

Project Research Questions

My main goal is to predict the outcomes of upcoming VCT matches, so the response variable is match win/loss, coded as 1 for a win and 0 for a loss. This is a classification problem because the outcome is binary. I expect predictors like team ACS, K/D ratio, first-kill rate, map, and tournament stage to be especially useful. The project is mostly predictive, but it also has a descriptive component because I want to understand which stats are most associated with winning.

Project Timeline

By the end of week 3, I plan to have the data cleaned, loaded, and merged into a workable format. In weeks 4 and 5, I will do exploratory data analysis, make charts, and identify important predictors and missing-data patterns. In weeks 6 and 7, I will build and compare classification models, then evaluate performance and refine the best one. In weeks 8 until the end of the quarter, I will interpret these results, and finalize my conclusions.

Questions

As of now my main consern is when using an extensive amount of CSV files, about 21, should i keep them separete or should I combine them all and clean the data from there? Other questions that arise I plan to go to office hours to ask!